FIITJEE

JEE(Main)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – IV
(Main)

S. No.	PHYSICS	S. No.	CHEMISTRY	S. No.	MATHEMATICS
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2.	**A**	32.	**A**	62.	**D**
3.	**A**	33.	**B**	63.	**B**
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6.	**B**	36.	**A**	66.	**A**
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20.	**B**	50.	**A**	80.	**C**
21.	**A**	51.	**D**	81.	**C**
22.	**C**	52.	**B**	82.	**C**
23.	**A**	53.	**D**	83.	**B**
24.	**D**	54.	**A**	84.	**A**
25.	**D**	55.	**B**	85.	**B**
26.	**D**	56.	**A**	86.	**C**
27.	**B**	57.	**C**	87.	**D**
28.	**C**	58.	**B**	88.	**B**
29.	**D**	59.	**A**	89.	**B**
30.	**D**	60.	**D**	90.	**A**

Physics

PART – I

SECTION – A

1. **B**

Population covered = $\rho\pi d^2$

$50\times10^5 = \rho(\pi hR)$

$\therefore \rho = \dfrac{50\times10^5}{3.14\times2\times150\times(6.4\times10^6)} = 828.6 \text{ km}^{-2}$

2. **A**

Fact about P – N junction.

3. **A**

Given that $v = v_o \log_e(r/r_o)$

field $E = \dfrac{-dv}{dr}$ or $E = -V_o\left(\dfrac{r_o}{r}\right)$

Now $eE = \dfrac{mv^2}{r}$ or $\dfrac{eV_or_o}{r} = \dfrac{mv_o^2}{r}$

$\therefore v = \left(\dfrac{ev_or_o}{m}\right)^{1/2}$

Now $mv = (mev_or_o)^{1/2} = \text{constant}$

$\therefore mvr = \dfrac{nh}{2\pi}$ Bohr's quantum condition

or $r \propto n$

4. **C**

$T = 2\pi\sqrt{I/MB_H}$ $\therefore T_1 = 2\pi\sqrt{\left[\dfrac{2I}{(P_m+2P_m)B_H}\right]}$

$T_2 = 2\pi\sqrt{\left[\dfrac{2I}{(2P_m+P_m)B_H}\right]}$

$\therefore \dfrac{T_1}{T_2} = \dfrac{1}{\sqrt{3}}$

5. **D**

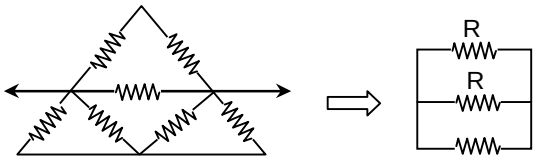
Let lengths of ABC and ADC is l_1 and l_2 respectively. Then

$\dfrac{l_1}{l_2} = \dfrac{360-60}{60} = 5$ Now $\dfrac{i_1}{i_2} = \dfrac{l_1}{l_2} = \therefore i_2 = 5i_1$

Now $B_1 = \dfrac{5}{6}\left(\dfrac{\mu_o i_1}{2r}\right)$ and $B_2 = \dfrac{1}{6}\left(\dfrac{\mu_o i_2}{2r}\right)$

$\therefore B_1 = B_2$

6. **B**



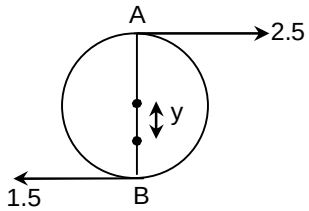
So $R_{eq} = 4R/9$

For Pmax, r = Reg. $\therefore r = 4R/9$

7. **B**

$$\frac{2.5}{2+y} = \frac{1.5}{2-y}$$

$\therefore y = 0.5$ m



8. **C**

$f_1 = f_2$

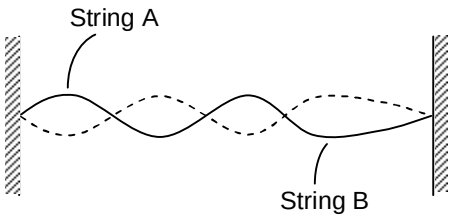
$D_1 < D_2$

$C = fD$

$\therefore C_1 < C_2$

but $C = \sqrt{\frac{T}{\mu}}$

$\therefore \mu_1 > \mu_2$



9. **C**

At the surface of a current carrying wire

$$B = \frac{1}{2}\mu_o Jr$$

$$J \propto \frac{B}{r}$$

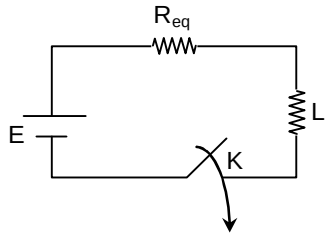
$\therefore$ as slope of wire P > slope of wire Q

10. **C**

Equivalent circuit

$$Req = R + \frac{2R \times 2R}{4R} = 2R$$

$\therefore$ time constant $t = \frac{L}{reg} = \frac{L}{2R}$



11. **D**

$\sum F_x = 0, \sum F_y = 0 \therefore \mu N = Mg$

$\sum \tau$ about O = 0

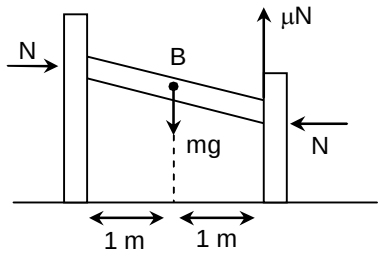
$\therefore \left[N\frac{l}{2}\sin\theta\right](2) = \mu N\frac{l}{2}\cos\theta$

$\therefore \tan\theta = \mu/2$

$\sec\theta = \left(\sqrt{\mu^2+4}\right)/2$

$\text{thus}\left(\frac{l}{2}/1\right) = \sqrt{\frac{\mu^2+4}{2}} \therefore l = \sqrt{\mu^2+4}$

$l = \sqrt{0.25+4} \therefore l = \frac{\sqrt{17}}{2}\text{m}$



12. **D**

$\frac{\mu_2}{v} - \frac{\mu_1}{v} = \frac{\mu_2 - \mu_1}{R}$

$\frac{1}{v} - \frac{1.5}{-u} = \frac{1-1.5}{-R}$

for v to be +ve, $\frac{1}{2R} - \frac{3}{2u} > 0$ or u > 3 R

13. **A**

N_B reaches maximum when activity of A and B will be same

i.e $D_A N_A = D_B N_B \therefore N_B = \frac{D_A N_A}{D_B}$

14. **B**

$V_B = 0 \therefore \frac{K_Q}{b} + \frac{K_{Q'}}{b} = 0$ or $Q' = -Q$

Where Q is total charge inside A and Q^1 is charge on shell B therefore field outside B will be zero and electric field inside A will be $E = \frac{KQ}{a^3}r$ (linearly increase) between spherical shell A and B $E = KQ/r^2$.

15. **C**

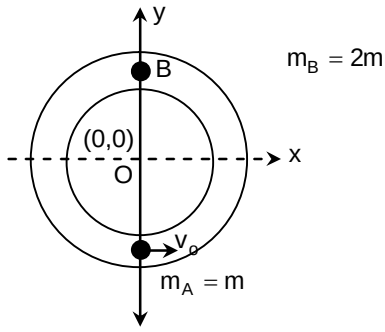
$-mv_o + o = mv_1 + 2mv_2$

$-v_o = v_1 + 2v_2 .....(i)$ & $e = \frac{v_2 - v_1}{-v_0} = 1$

$\therefore v_2 - v_1 = -v_o ....(ii)$ from (i) & $(ii) v_1 = \frac{v_o}{3}, v_2 = \frac{-2v_o}{3}$

$\therefore T = \frac{2\pi R}{v_o}$

Angle rotated by by A = $\frac{2\pi}{3}$, Angle rotated by B = $\frac{4\pi}{3}$rad.



16. **A**

Velocity of image of particle B,

$\vec{V}_B = 5i + 4(-i) = i$

Velocity of image of particle A,

$\vec{V}_A = 10(-i) + 4(-i) - 10j$

$\therefore \vec{V}_A = -14i - 10j$

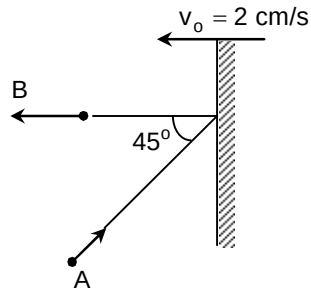
Relative velocity of image

$\vec{V}_{BA} = \vec{V}_B - \vec{V}_A$

$= i - [14(-i) - 10j]$

$= 15i + 10j$

$\therefore \vec{V}_{BA} = \sqrt{325}$ cm/sec



17. **D**

Since $P_1V_1 > P_2V_2 \Rightarrow T_A > T_B$ work done is +ve and efficiency is positive quantity

$\eta = \frac{\text{work done}}{\text{Heat given}} = 1 - \frac{Q_{released}}{Q_{absrobed}}$

Therefore there must be some heat absorbed in the process. During adiabatic process $\Delta Q = 0$, therefore during linear process heat will enter and leaves system at different times.

18. **A**

AV = const.

$P + \frac{1}{2}\rho v^2 = \text{constt}$

If velocity increases then pressure decreases.

19. **A**

By $\rho = m/v$ so percentage error can find.

20. **B**

$\left(\frac{dy}{dx}\right)_{tangent} = 2\cos\frac{\pi x}{L}$

and $\tan(90 + 45^o) = \left(\frac{dy}{dx}\right)_{normal}$

$\Rightarrow x = \frac{L}{3}, y = \frac{\sqrt{3}L}{\pi}$

21. **A**

Fact

22. **C**

$mvr = \frac{h}{2\pi}$ in first orbit

$v = \frac{h}{2\pi mr}$

$\therefore a = \frac{v^2}{r} = \frac{h^2}{4\pi^2 m^2 r^3}$

23. **A**

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}\therefore u=\frac{fv}{f-v}$$

$$\therefore \frac{v}{u}=\frac{f-v}{f}=1-\frac{v}{f}$$

$$|m|=\left|1-\frac{v}{f}\right|\therefore \text{slope}\frac{1}{f}=b/c$$

24. **D**

$$SL_2-SL_1=10\log\frac{I_{max}}{I_{min}}=10\log\left[\frac{\left(\sqrt{I_1}+\sqrt{I_2}\right)^2}{\left(\sqrt{I_1}-\sqrt{I_2}\right)^2}\right]$$

$$=10\log\frac{\left(\frac{a_1}{a_2}+1\right)^2}{\left(\frac{a_1}{a_2}-1\right)^2}=20\text{ dB}$$

25. **D**

$$a=\frac{iBl}{m}$$

$$\text{and } v=\left(\frac{iBl}{m}\right)t$$

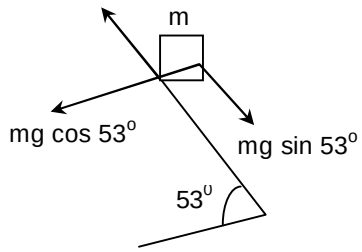
$$\therefore v=20\text{ cm/s}$$

26. **D**

$a=g\sin 53^o=\frac{4g}{5}$ time taken by block to reach the bottom

$$\therefore t=\sqrt{\frac{25}{a}}=\sqrt{\frac{25}{2\times 10}}=\sqrt{\frac{5}{4}}$$

$$\therefore \text{Distance travelled by the inclined}=vt=4\sqrt{\frac{5}{4}}=2\sqrt{5}\text{ m}$$



27. **B**

$$\text{speed } v=\sqrt{v_{x^2}+v_{y^2}}\therefore 2v\frac{dv}{dt}=2v_xa_x+2v_ya_y$$

$$\frac{dv}{dt}=\frac{v_xa_x+v_ya_y}{v}=\frac{3\times 2+4\times 1}{\sqrt{3^2+4^2}}=2\text{ m/s}^2$$

28. **C**

Displacement of the winning team is zero. Hence work done by loosing team on the winning team is zero.

Ground applies a frictional force on the losing team in a direction opposite to its displacement hence work done by ground on losing team is zero.

29. **D**

$$\int \vec{E}.\vec{dl} = -A.\frac{dB}{dt}$$

$$E(2\pi r) = -17\left[\frac{d}{dt}\{17 + 0.2\sin(\omega t + \phi)\}\right]$$

$$\therefore E = \frac{-r}{2}(0.2)\omega\cos(\omega t + \phi)$$

$\therefore$ magnitude of amplitude = $\frac{r}{2}(0.2) = 240$ m N/C

30. **D**

Zero in fact A will be inclined away from the vertical at some angle.

Chemistry

PART – II

SECTION – A

31. **A**
Factual

32. **A**
They make non ideal solution with negative deviation so $\Delta V_{mix} < 50$.

33. **B**
Initial nuclides of X and Y $\Rightarrow X_o$ and Y_o
After time 't' nuclides of X and Y $\Rightarrow X_t$ and Y_t
$X_t = 1/2Y_o, Y_t = 1/4X_o$
$X_o = 1.414Y_o, Y_t = 1/4X_o$
Now, $X_t = X_o e^{-D_1/t} = 1/2Y_o; Y_t = Y_o e^{-D_2/t} = 1/4Y_o$
Dividing equation (i) by (ii) ; we get
$X_o = \sqrt{2}Y_o \therefore 2 = 2e^{(D_2 - D_1)t}$
$\therefore e^{(D_2 - D_1)t} = 1$
Q since t is not zero so $D_2 = D_1 \Rightarrow$ 8 days.

34. **C**
For cyclic process du = 0
$q = -w$
$pv_1 = 1 \times R \times 300$
$pv_2 = 1 \times R \times 400$
$\therefore p(v_2 - v_1) = R(400 - 300)$
$q = -(w_{BA} + w_{AC} + w_{CB})$
$-600 = -[0 - 2(400 - 300) + w_{CB}]$
$w_{CB} = 800$ cal.

35. **D**
Anodic reaction –

$KOH \rightleftharpoons K^+ + OH^-$

$1 - 0.75 \quad 0.75 \quad 0.75$

$\left[H^+\right]_a = \dfrac{10^{-14}}{0.75}$

Cathodic reaction

$HCl \rightleftharpoons H^+ + Cl^-$

$0.1 \quad 0 \quad 0$

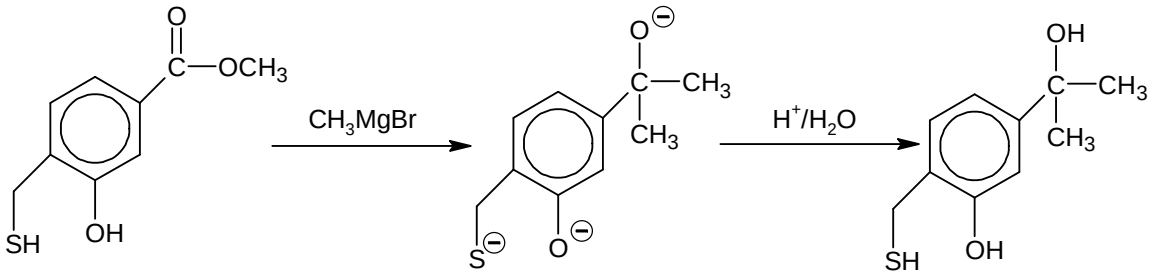
$0.1 \times 0.9 \quad 0.1 \times 0.9$

$\left(H^+\right)_C = 0.09$

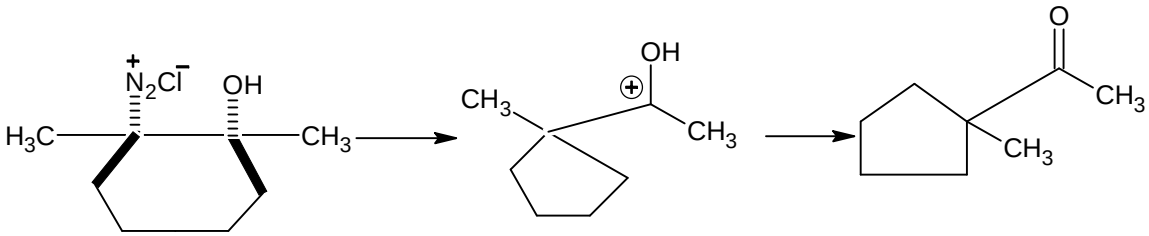
$$E_{cell} = 0 - \frac{0.0591}{2}\log\frac{\left[H^+\right]_a}{\left[H^+\right]_c} \quad \left[E^o_{cell} = 0,\ \text{it is concentration cell}\right]$$

on solving $E_{cell} = 0.378$ v

36. **A**



37. **A**



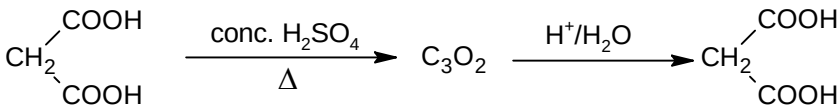
38. **B**

Witting Reaction, in presence of lithium salt trans alkene is formed.

39. **B**

'Al' is produce by electrolytic reduction of Al_2O_3

40. **D**



41. **D**

Conceptual.

42. **B**

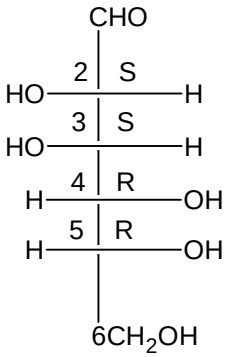
Conceptual.

43. **A**

$\left[t_{2g}^{2,2,1}\ell_g^{0,0}\right]$

44. **B**

C – 2 epimer of D – glucose



45. **C**

$XeF_2 + PF_5 \longrightarrow XeF^+ + PF_6^-$

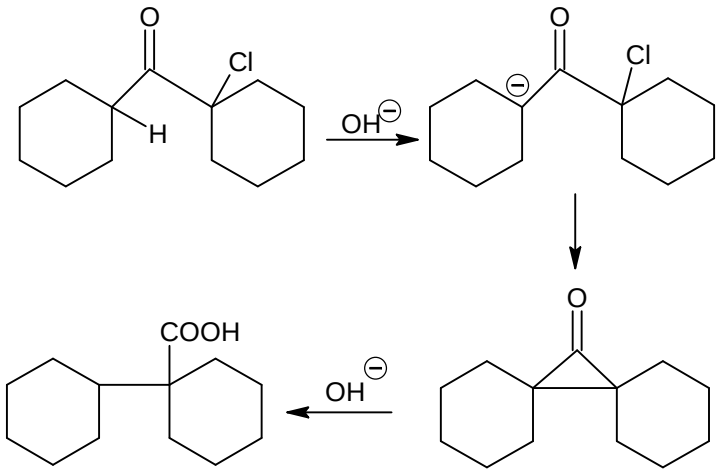
$XeF_4 + SbF_5 \longrightarrow XeF_3^+ + SbF_6^-$

46. **A**

Due to NGP reaction proceeds with retention of configuration.

47. **C**

48. **C**



49. **D**

Factual.

50. **A**

Initial moles of acetic acid $= \dfrac{0.5 \times 100}{1000} = 0.05$

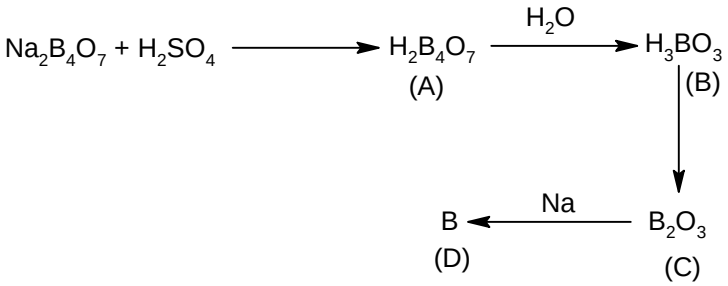
Moles of acetic acid left $= \dfrac{0.49 \times 100}{1000} = 0.049$

Moles of acetic acid absorbed $0.05 - 0.049 = 0.01$ mol in a litre

Molecules $6.022 \times 10^{23} \times 0.001 = 6.022 \times 10^{2}$ in 100 ml

Area occupied by single molecule of acetic acid $= \dfrac{\text{Total area}}{\text{no. of molecules}} = \dfrac{3.01 \times 10^{2}}{6.022 \times 10^{20}} = 5 \times 10^{-19}$ m^2

51. **D**



$\therefore$ Boron does not react with HCl, dil H_2SO_4 but react with conc. H_2SO_4 to form H_3BO_3

$$2B + 3H_2SO_4 \underset{(conc.)}{\longrightarrow} 2H_3BO_3 + 3SO_2$$

52. **B**

$$DE = -2.18 \times 10^{-18}\left[1-\frac{1}{4}\right]$$

$$= -2.18 \times 10^{-18} \times \frac{3}{4} J/atom$$

$$= -2.18 \times 10^{-18} \times \frac{3}{4} \times 6.022 \times 10^{20} \text{ Jouls for } \frac{1}{100} \text{moles}$$

$$= -2.18 \times \frac{3}{4} \times 6.022 \times 10^{2} \text{ J.}$$

$$= -3940 \text{ J}$$

Now for isothermal expansion $\Delta U = 0$

$\therefore$ dq = –dw

$$\text{so } w = -2.303\, nRT \log\frac{v_2}{v_1}$$

$$-39400 = -2.303 \times 2 \times 8.314 \times 300 \times \log\frac{v_2}{v_1}$$

$$\log\frac{v_2}{v_1} = \frac{39400}{11488.3}$$

$$\log\frac{v_2}{10} = 0.3$$

$$\text{or } \frac{v_2}{10} = 2$$

$v_2 = 20$ litre

53. **D**

Let no. of $Fe^{2+} = x$

no. of $Fe^{2+} = 0.93 - x$

then $2x + 3(0.93 - x) = 2$

$x = 0.79$

54. **A**

Solution will make buffer before equation point since before equation point both HA and NaA are present.

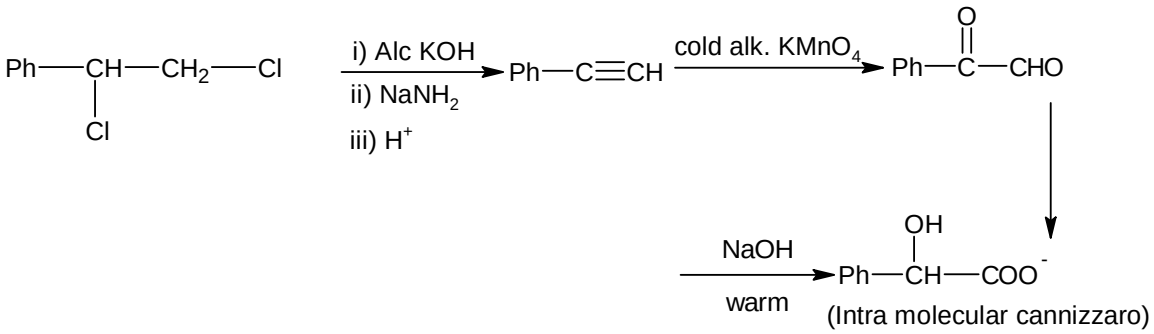
55. **B**
The Product is phenol – for aldehyde resin which is brownish mass with no alcoholic group.

56. **A**
Conceptual.

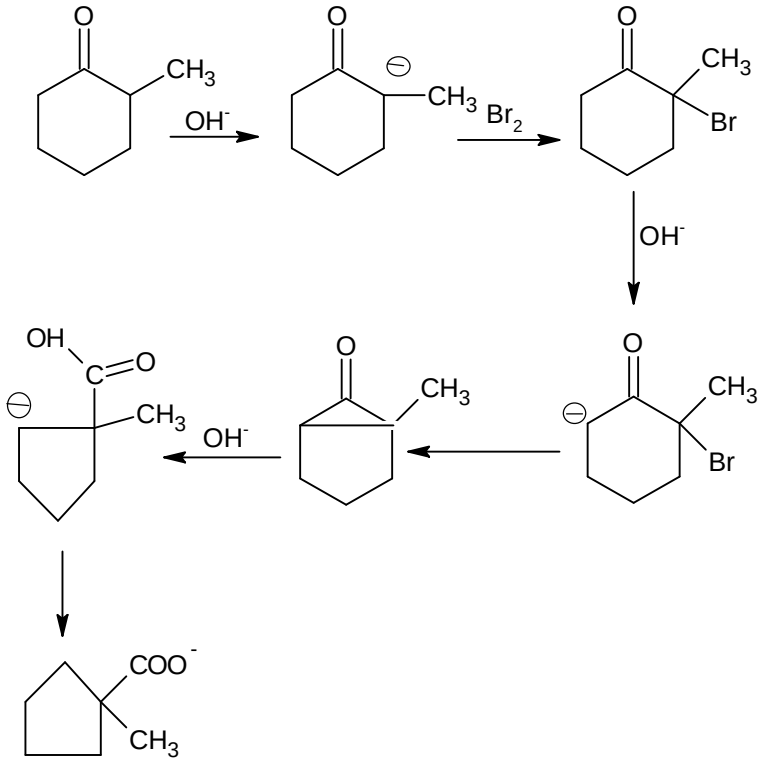
57. **C**
If no electronic repulsion is there than He can be treated as two mono electronic atoms so
$E_{He} = -13.6 \times \frac{z^2}{n^2} \Rightarrow -13.6 \times 4 = -54.4$ eV
For both electronic system $\Rightarrow -54.4 \times 2 = -108.8$ eV.

58. **B**
Conceptual.

59. **A**



60. **D**



Mathematics

PART – III

SECTION – A

61. **B**

Centre of rectangular hyperbola = (1, -2)
So equation of asymptotes are x = -1; y = -2
So radius of circle = $\frac{\sqrt{5}}{2}$

62. **D**

$\frac{1}{a}+\frac{1}{b}=\frac{1}{4} \Rightarrow b=\frac{4a}{a-4}$

so $A=\frac{4a^2}{2(a-4)}$

$A_{min} = 32 = c$
So last digit of $c^c = 6$

63. **B**

We have $k=1-\frac{2}{\alpha+1}-\alpha^2$
Gives k = -1 only

64. **C**

$t_2=-t_1-\frac{2}{t_1}$ also $t_1t_2=-4$

$\Rightarrow t_1=\pm\sqrt{2}$

so, $m_1 = \sqrt{2}$ and $m_2 = -\sqrt{2}$

$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|=2\sqrt{2}$

$\sin\theta=\frac{2\sqrt{2}}{3}$

so, $(f(\alpha))_{max}=\sqrt{\frac{8}{9}+8}=\frac{4\sqrt{5}}{3}$

65. **D**

$\begin{vmatrix}1 & 1 & -2\\ 2 & -1 & 1\\ p & q & -r\end{vmatrix}=0 \Rightarrow p+5q-3r=0$

so, $2px+3qy+\frac{4}{3}(p+5q)=0$

$p\left(2x+\frac{4}{3}\right)+q\left(3y+\frac{20}{3}\right)=0$

so fixed point is $\left(-\frac{2}{3},-\frac{20}{9}\right)$

so the slope of required line = $-\frac{3}{10}$

66. **A**

$a = 20 - 6\pi,\ b = 10\pi - 30,\ c = \sin^{-1}\sin(4\pi - 10) = 10 - 3\pi$

$f(x) = \sin(2(a+b+c)x) + \cos^2((a+b+c)x)$

$= \sin 2\pi x + \cos^2 \pi x$

$= \sin 2\pi x + \frac{1+\cos 2\pi x}{2}$

maximum value of $f(x) = \frac{\sqrt{5}+1}{2}$

67. **D**

Clearly $C \equiv (0,0)$, let the equation of circle be $x^2 + y^2 - ax - by = 0$.

On homogenizing with chord $11x + 3y - 14 = 0$

and comparing with pair of asymptotes we get equation of circle as

$x^2 + y^2 - \frac{112x}{23} - \frac{64y}{23} = 0$

68. **B**

Equation of hyperbola $(x-1)(y-2) = 4$

$\Rightarrow XY = 4$

If normal at $\left(ct, \frac{c}{t}\right)$ intersect curve again at $\left(ct', \frac{c}{t'}\right)$ then t' = $-\frac{1}{t^3}$

$2t = 4 \Rightarrow t = 2$

$(X', Y') = \left(-\frac{1}{4}, -16\right)$

$\Rightarrow (\alpha, \beta - 29) = \left(\frac{3}{4}, -14\right)$

$\Rightarrow \alpha = \frac{3}{4}, \beta = 15$

so $\frac{\beta}{\alpha} = \frac{15 \times 4}{3} = 20$

69. **A**

$2y\frac{dy}{dx} = 1$

$\Rightarrow \frac{dy}{dx} = \frac{1}{2y} = 1$

$\Rightarrow y = \frac{1}{2}$

so, $d = \sqrt{\frac{1}{16} + \frac{1}{16}} = \frac{1}{2\sqrt{2}}$

so $|\sin\alpha| = 1$

so number of solution is 3

70. **B**

The director circle of $\frac{x^2}{2}+\frac{y^2}{1}=1$ is $x^2+y^2=3$

Let $P\left(\left(\sqrt{3}\cos\theta,\sqrt{3}\sin\theta\right)\right)$

Chord of contact is $x\sqrt{3}\cos\theta+2y\sqrt{3}\sin\theta-2=0$

It touches $x^2+y^2=r^2$

$\Rightarrow r=\frac{2}{\sqrt{3\cos^2\theta+12\sin^2\theta}}=-\frac{2}{\sqrt{3+9\sin^2\theta}}$

$r_{max}=\frac{2}{\sqrt{3},}r_{min}=\frac{2}{\sqrt{12}}$

So, $\frac{A_{max}}{A_{min}}=4$

71. **B**

$x_1=2R\cos A,\ x_2=2R\cos B, x_3=2R\cos C$

$\frac{dx_1}{dA}=-2R\sin A$ also $a=2\sin A\Rightarrow\frac{da}{dA}=2R\cos A$

so, $\frac{dx_1}{da}=-\tan A,\frac{dx_2}{db}=-\tan B,\frac{dx_3}{dc}=-\tan C$

$\tan A+\tan B+\tan C\geq 3\sqrt{3}$

So, $\left(\frac{dx_1}{da}+\frac{dx_2}{db}+\frac{dx_3}{dc}\right)+11\leq 2$

72. **C**

Let $b+c=x$ and $bc=y$ and $a=z$

Then $(b+c+a)(b+c-a)=bc$

$\Rightarrow\sqrt{\frac{s(s-a)}{bc}}=\frac{1}{2}\Rightarrow\cos\frac{A}{2}=\cos\frac{\pi}{3}\Rightarrow A=120^\circ.$

$r=\frac{\Delta}{s}=\frac{\frac{1}{2}bc\sin A}{\frac{a+b+c}{2}}=\frac{y\sin 120^\circ}{\frac{x+y}{2}}=\frac{y\sqrt{3}}{x+z}$

73. **C**

Let $\frac{x^2}{a_n^2}+\frac{y^2}{b_n^2}=1, a_n>b_n$ be the ellipse with eccentricity = e_n

$\Rightarrow b_n^2=a_n^2\left(1-e_n^2\right)$ .....(1)

According to question $b_n=b_{n-1}$ ....(2)

And $a_{n-1}=a_ne_n$ .....(3)

Again for ellipse $a_n^2e_n^2=a_n^2\left(1-e_n^2\right)\left(1-e_{n-1}^2\right)$ ....(4)

Solving we have $a_n^2e_n^2=a_n^2\left(1-e_n^2\right)\left(1-e_{n-1}^2\right)$

$\Rightarrow e_n^2=\left(1-e_n^2\right)\left(1-e_{n-1}^2\right)$ ...(5)

eccentricity e_n is independent of $n \Rightarrow e_n = e_{n-1} = e$

$e^2 = \left(1 - e^2\right)^2 \Rightarrow e = \dfrac{\sqrt{5}-1}{2}$

74. **C**

$\tan\left((\theta°) + (30-\theta)°\right)$

$= \dfrac{\tan\theta° + \tan(30-\theta)°}{1-(\tan\theta°)(\tan(30-\theta)°)} = \dfrac{1}{\sqrt{3}} \Rightarrow \sqrt{3}\left(\tan\theta° + \tan(30-\theta)°\right) = 1-(\tan\theta°)(\tan(30-\theta)°)$

$\left(\sqrt{3} + \tan\theta°\right)\left(\sqrt{3} + \tan(30-\theta)°\right)$

$= 4^{14}\left(\sqrt{3} + \tan 15°\right) = 4^{14}\left(\sqrt{3} + \left(2-\sqrt{3}\right)\right) = 2^{29}$

75. **D**

$f(x) = 1 - x + x^2 - x^3 + ... + x^{16} - x^{17} = \alpha_0 + \alpha_1(1+x) + \alpha_2(1+x)^2 + ... + \alpha_{17}(1+x)^{17}$

$\Rightarrow f''(-1) = 2.1 + 3.2 + 4.3 + ... + 17.16 = 2.1\alpha_2$

$\Rightarrow 2\alpha_2 = \sum_{r=1}^{16} r(r+1) = \dfrac{1}{3}\sum_{r=1}^{16}\left(r(r+1)(r+2) - (r-1)r(r+1)\right) = \dfrac{1}{3}(16.17.18)$

$\Rightarrow \alpha_2 = 816$

76. **B**

We have $f(x+1) = \dfrac{f(x)-1}{f(x)+1}$. For convenience, let $f(x) = \alpha_x$

We have $\alpha_{n+1} = \dfrac{\alpha_n - 1}{\alpha_n + 1}$

Thus $\alpha_{n+2} = \dfrac{\alpha_{n+1}-1}{\alpha_{n+1}+1} = \dfrac{\alpha_n - 2}{2\alpha_n}$

Continuing on, we have $\alpha_{n+3} = \dfrac{\alpha_{n+2}-1}{\alpha_{n+2}+1} = \dfrac{-\alpha_n - 2}{3\alpha_n - 2}$

$\alpha_{n+4} = \dfrac{\alpha_{n+3}-1}{\alpha_{n+3}+1} = \dfrac{-4\alpha_n}{-4} = \alpha_n$

Thus the function is periodic with length 4

77. **A**

$xy' - 2y = x^4y^2, y(1) = -6$

Divide by x: $y' - \dfrac{2y}{x} = x^3y^2$

Let $y = u^{-1}, \frac{dy}{dx} = -u^{-2}\frac{du}{dx}$ chain rule

This gives: $-u^{-2}\frac{du}{dx} - \frac{2}{x}u^{-1} = x^3u^{-2}$

$$\frac{du}{dx} + \frac{2}{x}u = -x^3$$

The integrating factor is $e^{\int \frac{2}{x}dx} = x^2$

But, $u = y^{-1}$, therefore: $y = \frac{-6x^2}{x^6 - 6C}$

So $y = \frac{-6}{x^4}$

78. **B**

Let $I = \int_0^{2\alpha} \frac{\sqrt[4]{\sin(3\alpha - x)}}{\sqrt[4]{\sin(3\alpha - x)} + \sqrt[4]{\sin(\alpha + x)}}dx$

$$= \int_0^{\alpha} \frac{\sqrt[4]{\sin(3\alpha - x)}}{\sqrt[4]{\sin(3\alpha - x)} + \sqrt[4]{\sin(\alpha + x)}} + \frac{\sqrt[4]{\sin(3\alpha - (2\alpha - x))}}{\sqrt[4]{\sin(3\alpha - (2\alpha - x))} + \sqrt[4]{\sin(\alpha + (2\alpha - x))}}dx$$

$$\Rightarrow I = \int_0^{\alpha} dx = \alpha$$

79. **D**

$$\frac{2a}{\tan A} = \frac{b}{\tan B}$$

$$\frac{2a\cos A}{\sin A} = \frac{b\cos B}{\sin B}$$

$2\cos A = \cos B$, by the sine rule

Therefore,

$$\frac{\sin^2 A - \sin^2 B}{\cos^2 A + \cos^2 B} = \frac{(1 - \cos^2 A) - (1 - \cos^2 B)}{\cos^2 A + \cos^2 B}$$

$$= \frac{1 - \cos^2 A - 1 + 4\cos^2 A}{\cos^2 A + 4\cos^2 B} = \frac{3\cos^2 A}{5\cos^2 A} = \frac{3}{5}$$

80. **C**

Let $z = x + iy$

$$3(x - iy) + 2\sqrt{x^2 + (y + 2)^2} + 9i = 0$$

$\Rightarrow y = 3$ and $3x + 2\sqrt{x^2 + (y+2)^2} + 9i = 0$

$9x^2 = 4x^2 + 100$

$x^2 = 20$

$x = \sqrt{20}$ $(\text{as } |\arg z| < \pi/2)$

so $z = \sqrt{20} + 3i$

In $\sqrt{20} + 3i = e^a.e^{ib}$

$e^a = \sqrt{29} \Rightarrow a = \ln\sqrt{29}$

81. **C**

$P^3 = P(I-P) = P - P^2 = P - (I-P) = 2P - I$

$P^4 = P(2P-I) = 2(I-P) - P = 21 - 3P$

$P^5 = P(2I-3P) = 2P - 3(IP) = 5P - 31$

$P^6 = P(5P-3I) = P(I-P) - 3P = 5I - 8P$

82. **C**

$$\left(1+\frac{1}{x}+\frac{1}{x^2}+\frac{1}{x^3}\right)^{10}\frac{\left(1+x+x^3+x^4\right)^{10}}{x^{30}}$$

coefficient of x^{30} in $\left(1+x+x^3+x^4\right)$

$= {}^{10}C_{10}\,{}^{10}C_0 + {}^{10}C_9 + {}^{10}C_3 + {}^{10}C_8\,{}^{10}C_6 + {}^{10}C_7\,{}^{10}C_9$

$= 1 + 1200 + 9450 + 1200$

$= 11851$

last digit of $(11853)^{11851} = 7$

83. **B**

$$n(S) = \frac{7!}{(2!)^2}$$

$$n(A) = \frac{5!3!}{7!} = \frac{1}{7}$$

84. **A**

$$\vec{\alpha}.\vec{\beta} = \frac{b}{a} + \frac{4a}{b} + 1$$

as $\frac{b}{a} + \frac{4a}{b} + 1 \geq 5$

so $\left(\frac{10}{5+\vec{\alpha}.\vec{\beta}}\right)_{max} = 1$

85. **B**

$$\frac{2\sqrt{2}}{2\pi} < \int_{\pi}^{2\pi} \frac{|\sin x + \cos x|}{x}\,dx < \frac{2\sqrt{2}}{2\pi} \qquad \ldots\ldots (i)$$

$$\pi < x < 2\pi \Rightarrow \frac{1}{2\pi} < \frac{1}{\pi} < \frac{1}{\pi}$$

$$\frac{2\sqrt{2}}{3\pi} < \int_{2\pi}^{3\pi} \frac{|\sin x + \cos x|}{x} dx < \frac{2\sqrt{2}}{2\pi} \qquad \text{.....(ii)}$$

adding (i) and (ii)

$$\frac{5\sqrt{2}}{3\pi} < AM \frac{3\sqrt{2}}{\pi} 0.75 < A < 1.3$$

86. **C**

As f(1) = 0

$f\left(1^+\right) > 0$

$f\left(1^+\right) < 0$

$\Rightarrow$ no local extremum at x = 1

87. **D**

$$I = xf(x)I_0^2 - \int_0^2 f(x)dx$$

$$= 0 - \frac{3}{4} = -\frac{3}{4}$$

88. **B**

$$\frac{\sin(x_n - x_{n+1})}{\sin x_n \sin_{n+1}} = \frac{1}{3^{\frac{2n+1}{2}}} \Rightarrow \sum_{n=1}^{\infty} \cot x_{n+1} - \cot x_n = \sum_{n=1}^{\infty} \frac{1}{3^{\frac{2n-1}{2}}}$$

$$\Rightarrow \lim_{n\to\infty} \cot x_n - \cot x_1 = \frac{\sqrt{3}}{2}$$

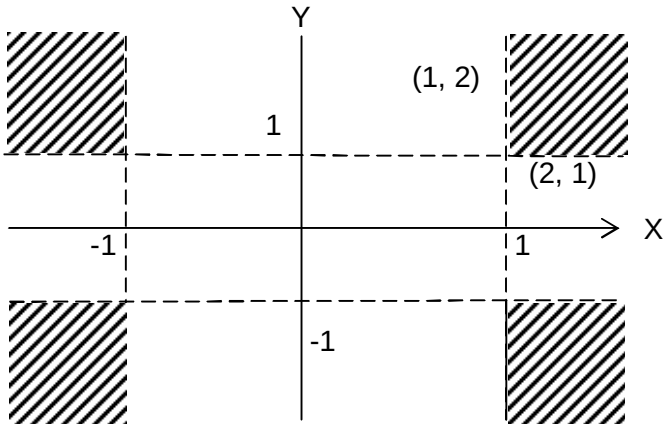
$$\Rightarrow \lim_{n\to\infty} x_n = \frac{\pi}{6} \text{ as } \cot x_1 = \frac{\sqrt{3}}{2}$$

89. **B**

Shaded region depicts min $(|x|, |y|) \geq 1$

Required area = $4\int_1^2 \left(\sqrt{5 - x^2} - 1\right) dx$

$$= 10\left(\sin^{-1}\frac{2}{\sqrt{5}} - \sin^{-1}\frac{1}{\sqrt{5}}\right) - 4$$



90. **A**

$$\int_{\pi/4}^{\pi/2} (\ell n(\cos x))e^x\left(\cot x - \operatorname{cosec}^2 x\right)dx = \left[\ell n(\cos x).e^x \cot x + e^x\right]_{\pi/4}^{\pi/2}$$

$$= \left[e^x\left(\ell n(\cos x).\cot x + 1\right)\right]_{\frac{\pi}{4}}^{\frac{\pi}{2}} = e^{\frac{\pi}{2}} - e^{\frac{\pi}{4}}\left(\ell n\frac{1}{\sqrt{2}} + 1\right)$$

$$= e^{\pi/4}\left[e^{\pi/4} + \frac{1}{2}\ell n2 - 1\right]$$